

gemsweep: a streamlined workflow for demixing metagenomic sample read sets to the strain level

Claudia Carter*, Sahra Musse*, Tom Morrison*, Will Roberts-Sengier*, Florent Lassalle* and Lewis Fisher*

*Parasites and Microbes Programme, Wellcome Sanger Institute



Background

Why use this workflow? Metagenomic sequencing is an efficient way to explore the composition of a sample, however, demixing co-sequenced genomes represents a significant challenge. *De novo* assembly-based metagenomic binning tools, such as MetaBAT2, are effective at the species-level yet collapse multiple strains due to their near identical composition and coverage profiles. Reference-based toolset mSWEEP [1] and mGEMS [2] probabilistically assign and taxonomically bin reads using pseudoalignment against reference genomes that are pre-clustered into strains. Binned reads are typically assembled downstream, and although subsequent analyses vary by research question, these tools represent a widely used core workflow.

What are the current problems with its use? Manual and sequential tool use is error prone, inefficient and lacks reproducibility. Optimisation requires detailed knowledge of compute resource requirements.

Also, users must first generate appropriate reference sets and provide a biologically-meaningful grouping. This is a key factor in the accuracy of mSWEEP relative abundance estimation; insufficient lineage representation in the references can cause false positive read to bin assignments [3] whilst strain-resolution results depend on clustering to the strain level. Suitable reference sets vary considerably by sample type and research question, from limited diversity in plate sweep metagenomes to broad environmental communities, and from species to clonal lineage resolution. No tool currently automates reference selection and preparation across this range of taxonomic scopes and granularities.

Here we present **gemsweep**, a pipeline that addresses these problems.

Objectives

- ❖ Automate workflow for scalability, reproducibility and portability
- ❖ Apply a heuristic for refining large reference sets to smaller representative sets
- ❖ Extend applicability to further metagenomic samples by automating selection of appropriate references
- ❖ Minimize time and computational cost of the pipeline

Automating and validating the core workflow

Implementation

- Modular Nextflow processes in Docker containers form a pipeline that is portable, scalable and reproducible
- Automated resource allocation with per-task escalation on failure, minimises need for user intervention
- Covers reference preprocessing: Themisto indexing and PopPUNK clustering
- Dedicated endpoint to the core workflow skips non-deterministic PopPUNK stage on repeat runs that require the same reference sets, preserving reproducibility

Validation

- Replicated original mGEMS publication [2] results using 3 synthetic mixed read sets against 218 *Escherichia coli* reference genomes
- Synthetic communities constructed for *Vibrio cholerae*, *E. coli*, and *Streptococcus pneumoniae* exploring a range of within-sample genomic relatedness
- Increased read-level noise observed in bins from closely related strains which is consistent with a known mSWEEP limitation
- mSWEEP relative abundance estimates benchmarked against comparable tool InStrain; strong concordance observed between tools (see Figure A)

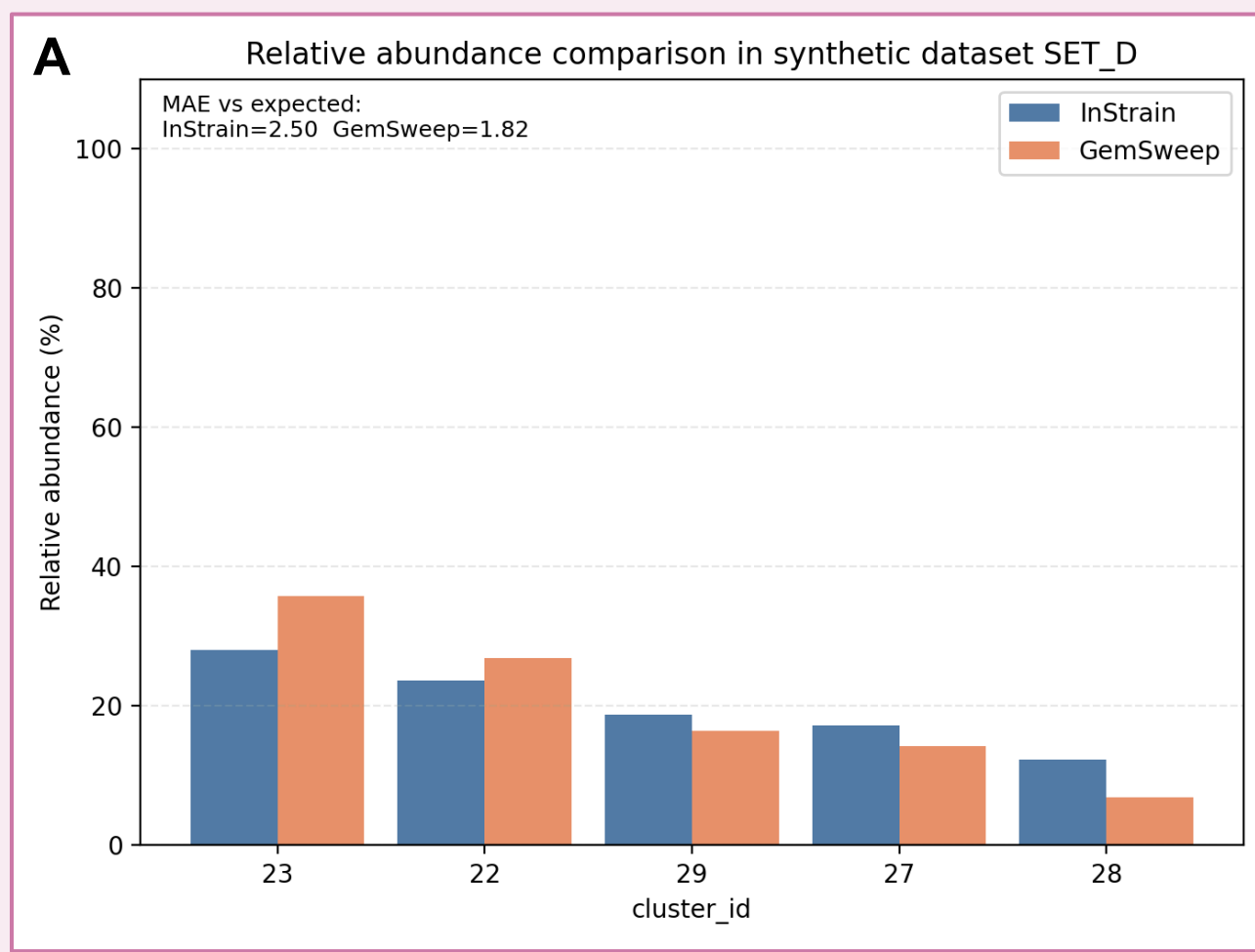
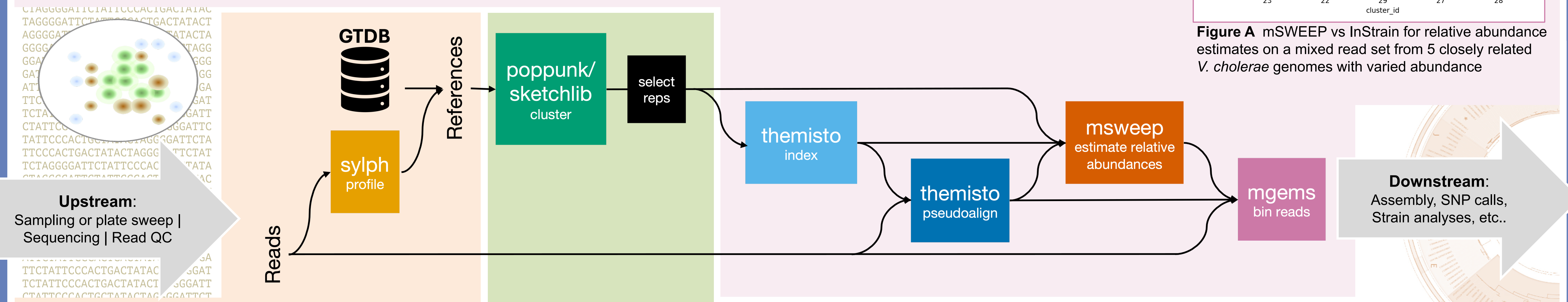


Figure A mSWEEP vs InStrain for relative abundance estimates on a mixed read set from 5 closely related *V. cholerae* genomes with varied abundance



Reference Autoselection

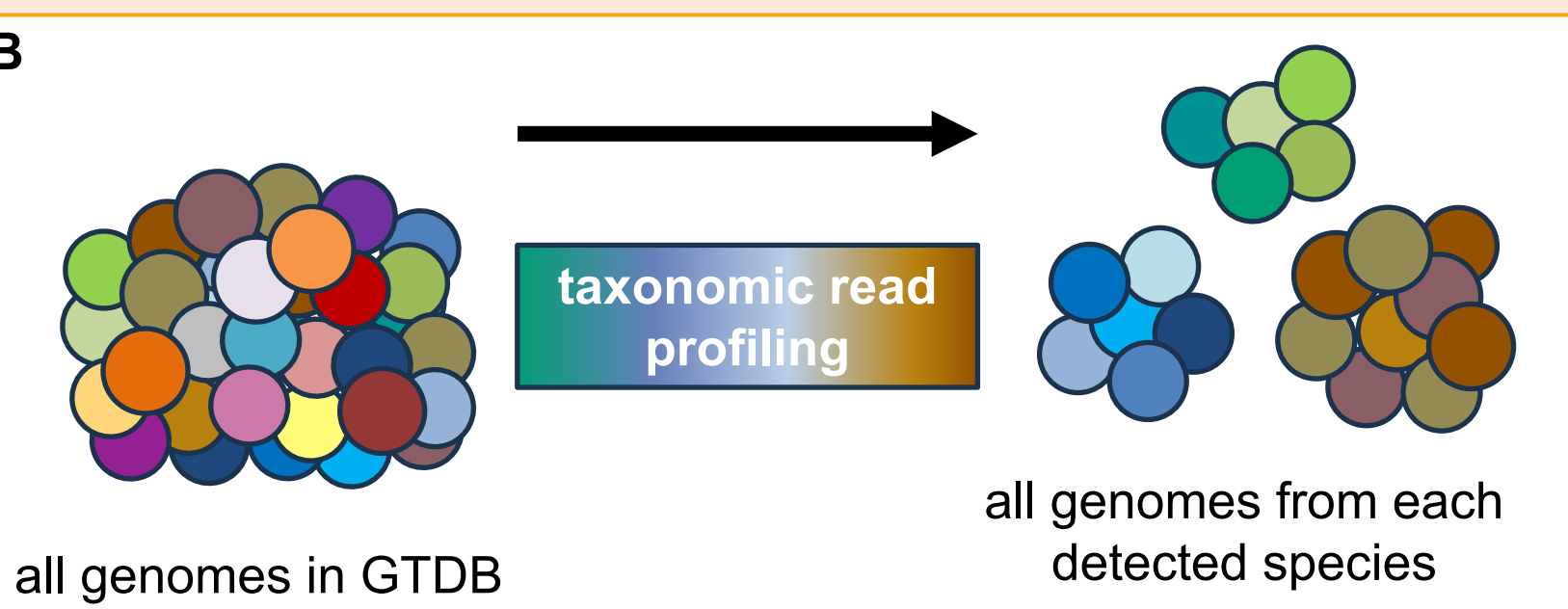


Figure B Reference selection; genomic variation indicated by shade but grouped by species (colour)

- A read profiling stage enables sample-guided reference selection
- All genomes for the detected species are pre-selected from the full GTDB [4] (release 226: >715,000 genomes)
- Genomes are clustered to strain level in per-species groups and further refined (see green section) to produce the reference set
- By this method reference sets are tailored to the sample, yet pre-selection at the species level ensures analysis is consistent across batches/ microbial communities
- Autoselection of references expands the use case for this workflow beyond plate sweeps to samples without prior knowledge of the present taxa
- To validate we use the Toy Human Gut samples of the CAMI dataset [5] – a synthetic community of known composition used to validate several metagenomic read binning methodologies

Reference Refinement

- Necessitated by the autoselection workflow but also useful for slimming down broad targeted reference sets for computational efficiency, we implemented (optional) ANI-based reference refinement
- Each cluster is capped to a number of representative genomes (*n*), which are maximally distant by ANI. This is done by a network-based trim; removing one genome out of each closest pair until only *n* representative genomes remain, effectively keeping the mutually most distantly related genomes
- This approach intentionally preserves within-cluster diversity – which is important for mSWEEP relative abundance estimation – yet cuts redundancy and reference set size, with savings in intermediate file size and computational cost particularly for pseudoalignment, and with a knock-on downstream effect
- We validated this method on two reference sets, capping clusters at 20 representatives:

Set A - 218 closely related *E. coli* genomes from the mGEMS release paper [2]
Set B - 14,000 diverse *E. coli* genomes from a study using this workflow [6]

	Set A	Set B
Reduction in # References	-64.7%	-81.0%
# Reduced / Total Clusters	1 / 8	68 / 425
Reduction in Combined Index File Size	-23.3%	-85.3%
Index Build Peak RAM	-23.2%	-78%
Reduction in wchar for Pseudoalignment	-58.9%	-91.7%
mSWEEP Peak RAM	-63.9%	-57%
mGEMS Peak RAM	-56.0%	-84%

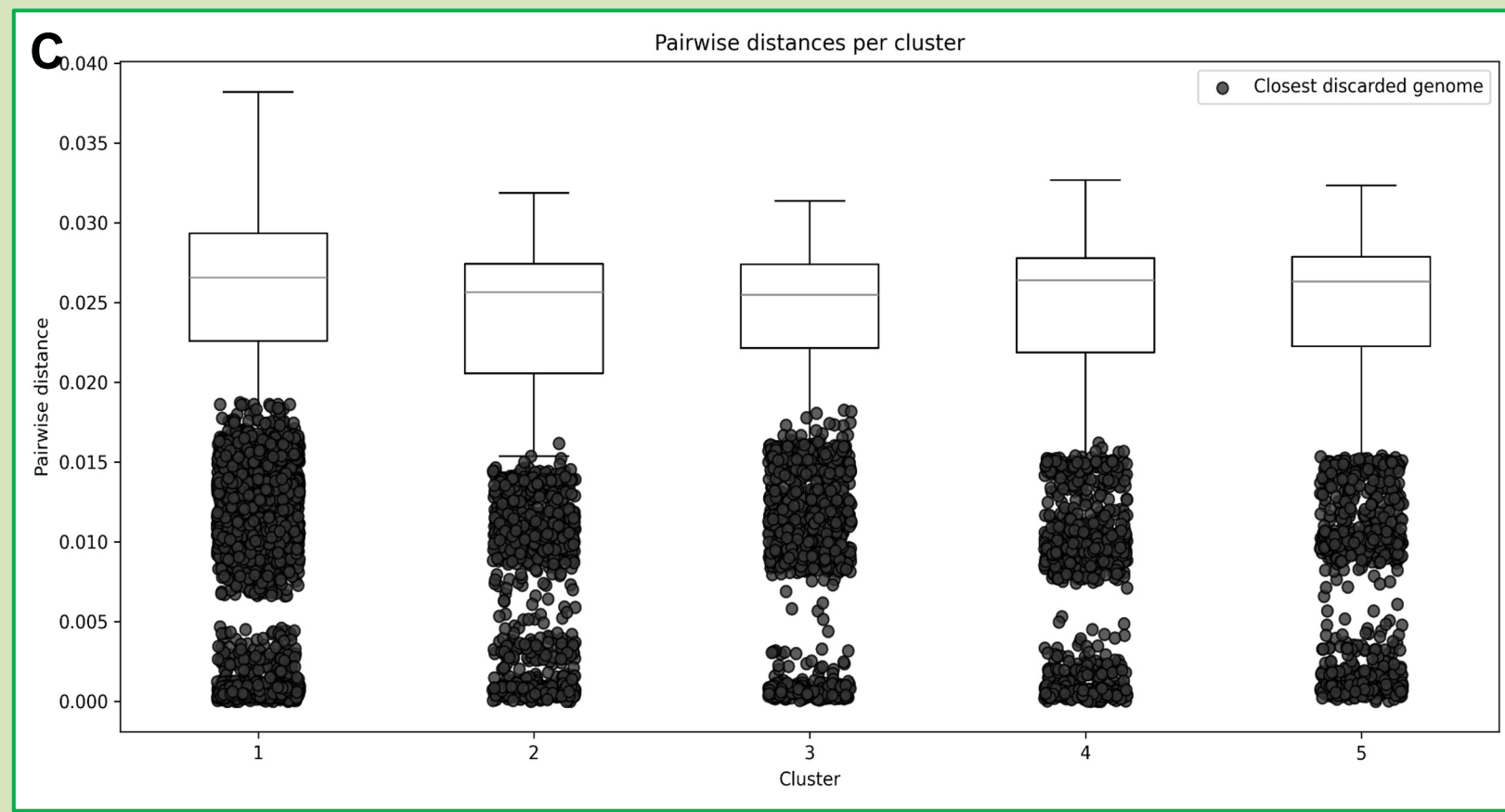
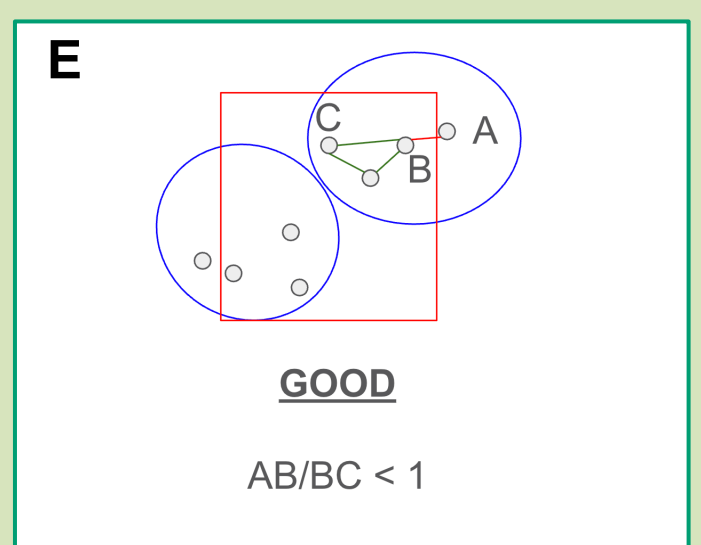
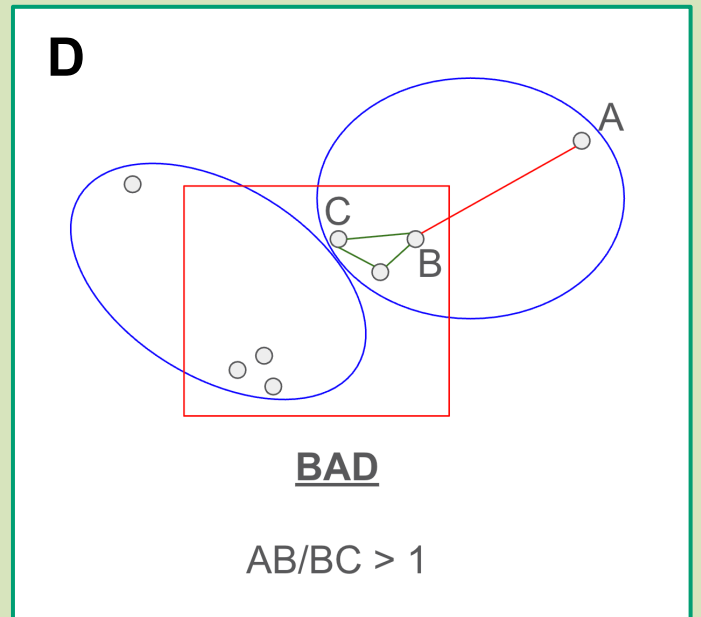


Figure C Per cluster pairwise distances for genomes in Set B, shown as a boxplot with distances to removed genomes plotted on top as dots – note whilst this figure illustrates just the 5 largest clusters, smaller clusters with >20 genomes show the same trend and the rest remain untouched

Table The effect of reference refinement on intermediate output written (wchar) and peak RAM on different ref sets



Figures D and E Clusters (blue circles) from which representative genomes (red box) are selected. Genome A is rightfully discarded in E and wrongfully discarded in D

Conclusion

We show here that **gemsweep** streamlines and builds upon a common workflow for the operationally demanding task of taxonomic binning of mixed microbial samples. By automating reference preparation and introducing options for reference refinement and sample-adaptive reference genome dataset selection, **gemsweep** substantially lowers the barriers to strain-resolved metagenomic analysis. The ANI-based reference refinement module balances computational tractability with biological representativeness, ensuring that reference clusters remain informative without being redundant. The Sylph-driven autoselection mode further removes the requirement for pre-curated reference databases, enabling the pipeline to be applied to most bacterial metagenomes.

Built in Nextflow, **gemsweep** is portable across HPC, cloud, and local environments, and is containerised for full reproducibility. The pipeline will be released as open-source software, and we invite the community to adopt and feedback for its continued development.

Future Directions & Limitations

- ❖ Release open source on github.com/sanger-pathogens
- ❖ GPU enabling (optional, for accessibility)
- ❖ Option to chain with our assembly pipeline
- ❖ Collaboration to adapt mSWEEP and mGEMS for viral analysis
- ❖ Currently reference autoselection only supports reference genomes in GTDB
- ❖ Plans to explore current and potential support for long read data

References

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